

A multi-timeframe liquidity-fade strategy on FX, gold and the Nasdaq

What it trades, why the effect exists, what it wins — and why that is still not enough

This document describes a systematic intraday strategy built and tested on two years of five-minute data for EUR/USD, USD/JPY, XAU/USD and the Nasdaq 100 — roughly 577,000 bars per instrument, with the bid *and* ask sides downloaded so that the spread is measured rather than assumed. It sets out the rule, the mechanism the rule is trying to exploit, the win rate it achieves, and the arithmetic that decides whether that win rate is worth anything.

The headline

The strategy wins **36.58%** of its resolved trades at a 2:1 reward-to-risk ratio. A no-information strategy with the same barriers wins **33.33%**. The gap is **$z = +3.85$** over 3,119 trades — a real effect, not noise.

It is worth **+0.100R per trade** gross. The measured spread costs **0.068R**, leaving **+0.032R** net — positive, but at **$t = +1.25$** , not statistically distinguishable from zero.

Add **0.35bp per side** of commission, an ordinary retail rate, and it becomes **−0.109R** at **$t = −4.21$** . **The strategy is not tradeable at retail cost.**

1 What the strategy does

The rule is a cascade. Each timeframe has exactly one job, and no timeframe is allowed to do another's.

Timeframe	Job	What it does
30 min	Find the setup	Detects a liquidity sweep (price pierces a 20-bar extreme and closes back inside it) or a retracement into a fair value gap.
5 min	Trigger the entry	First bar that closes beyond the previous 5-minute close, in the setup's direction. Fill at that close.
30 min	Manage the exit	Stop at 1.5 five-minute sigmas beyond the trigger bar's extreme; target at 2× the stop; time limit 20 bars.

Both directions are traded. Longs are 51.4% of the sample — the rule has no directional bias built into it, and none was found in the data.

Two further restrictions, both stated before the results were seen

Sessions. Each instrument trades only when its own order flow is live: USD/JPY in Tokyo and New York, EUR/USD and XAU/USD in London and New York, NAS100 in New York alone. Windows are cash hours in each centre's own clock, because London and New York shift their clocks three weeks apart each spring and Tokyo not at all.

Rollover blackout. No entries between 16:55 and 18:00 New York. The measured spread at the value-date roll is 9.3× its daily median on USD/JPY and 6.5× on EUR/USD.

These are liquidity claims, not fitted parameters — that distinction is what keeps them admissible. They could have been written down before a single backtest was run, and they were.

2 Why the effect exists

The strategy fades failed extremes. Both setups are the same idea in different clothing.

The liquidity sweep

Resting stop orders cluster just beyond obvious highs and lows. When price reaches through such a level, those stops fire as market orders and briefly push price further — then, with the pool exhausted and no genuine demand behind the move, price returns inside the range. The setup requires exactly that shape: a pierce of the 20-bar extreme, and a close back inside it. The trade is taken against the pierce.

The fair value gap

A fair value gap is a three-bar pattern where the middle bar moves so fast that bar one and bar three do not overlap — a price range through which almost nothing traded. The retracement setup requires the pullback to reach *into* that range, not merely to reach a given depth. This distinction turned out to matter a great deal, and section 5 quantifies it.

The reason to believe the mechanism

It is not the story. Stories about order flow are cheap and mostly unfalsifiable. The reason is that an independent study on the same instruments — a rank-information-coefficient screen over eight candidate signals, corrected for multiple comparisons and confirmed on a held-out period — found **short-horizon mean reversion and no momentum whatsoever**. Twelve of twenty-four tests survived correction in discovery and all twelve confirmed out of sample.

A fade is pointed the same way as the one effect that independently replicates in this data. That is a far better reason to hold a setup than any narrative about where the stops sit.

The corollary, which cost the strategy its original core

The breakout setup trades *with* the move, against the only measured effect. Its gross edge is **+0.049R**, less than half the two fades (+0.104R and +0.118R), and it loses **-0.039R** net. It was removed.

The original rule was built around a 9-period EMA of the session VWAP as a trend filter. Tested in isolation it removed two thirds of all trades and the win rate went *up* without it (35.62% against 35.11%). It was removed too.

3 The win rate

A win rate in isolation means nothing without the reward-to-risk ratio it was earned at. At 2:1, a driftless random walk hits its target before its stop **one time in three**: in log space the probability of touching a barrier b above before one a below is $b/(a+b)$, which for a target twice as far as the stop is exactly 1/3. So 33.33% is what a strategy with no information looks like, and it is the only honest comparison.

Each trade's own benchmark is computed from its own barriers and the realised win rate is tested against the sum with a Poisson-binomial z-statistic.

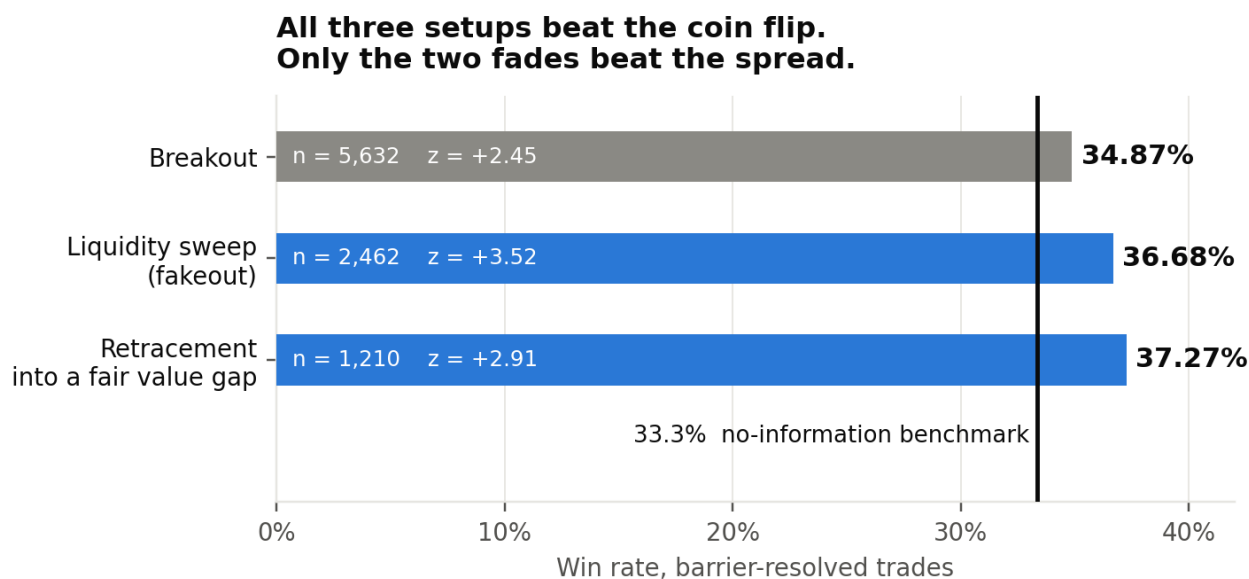


Figure 1. Win rate by setup type at a 1.5-sigma stop, 2:1 reward-to-risk, across all four instruments. The black line is the Black–Scholes no-information rate. Bars show barrier-resolved trades only; time exits are excluded because they pay neither +2R nor −1R and would flatter the comparison.

The strategy as it finally stands

Two fade setups, gold excluded, 1.5-sigma stop:

	Trades	Win rate	Benchmark	z	Net R	t
Overall	3,119	36.58%	33.33%	+3.85	+0.032	+1.25
Liquidity sweep	2,057	36.32%	33.34%	+2.87	+0.018	
Retracement + FVG	1,062	37.10%	33.33%	+2.60	+0.030	
EUR/USD	1,144	37.06%	33.33%		+0.025	
USD/JPY	1,203	37.41%	33.33%		+0.037	
NAS100	803	36.36%	33.33%		+0.035	

All three instruments are positive. Both setups beat the benchmark independently. Net R is gross of commission and net of the measured spread.

4 Why it still does not pay

Cost is charged from the **measured** spread, not an assumption. The ask side was downloaded alongside the bid, and ask minus bid over the mid is the spread as quoted. Median, basis points per side:

EUR/USD	USD/JPY	NAS100	XAU/USD
0.34 bp	0.34 bp	0.53 bp	1.62 bp

Gold is excluded on this number alone. Its spread is roughly five times the majors' and it loses −0.058R per trade despite having a perfectly respectable gross edge of +0.090R. The cost is simply larger than the edge, every time.

The deeper problem is that the edge and the cost move together. A tighter stop cuts the dollar loss per trade, but cost in units of risk is $2 \times \text{spread} \div \text{stop distance}$, so a tighter stop *raises* the hurdle. And the reversion being traded

lives within a few five-minute sigmas of the extreme — widen the stop to save on cost and the edge itself disappears.

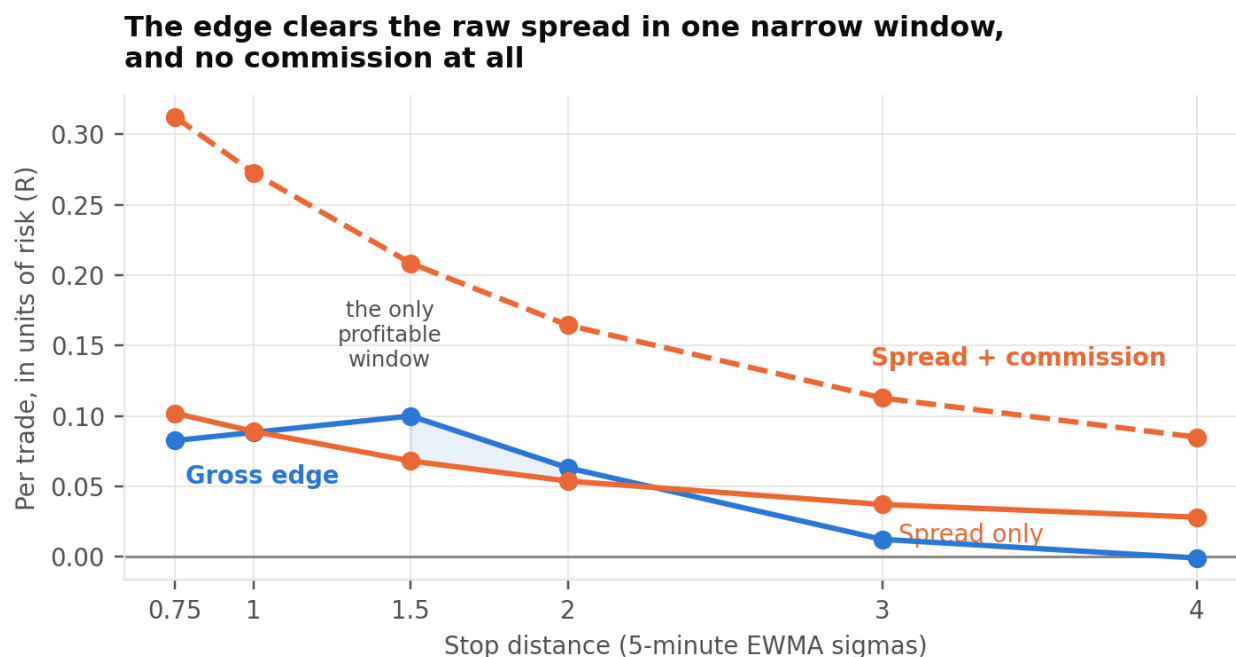


Figure 2. Gross edge and cost per trade, both in units of risk, across stop widths. The blue line clears the solid orange line only between roughly 1 and 2 sigmas — the shaded sliver. It never clears the dashed line, which is the same spread plus 0.35bp per side of commission. By 3 sigmas the gross edge is gone entirely.

What would have to change

The edge would need to be roughly **3×** larger, or the round-trip cost **two thirds** smaller, for this to clear a realistic retail cost base.

Neither is available at 1,600 trades a year. The binding constraint is not the entry rule, the stop, the timeframe or the filters — it is **frequency**. A 10bp effect cannot be harvested through a 7bp door sixteen hundred times a year.

Slippage is **not modelled at all**. The spread is what you are quoted, not what you get on a market order into a thin book — and the trades this rule likes are precisely the ones where those differ. Every figure here is therefore optimistic.

5 What was tested and discarded

Roughly 120 configurations were measured. The ones that failed are more informative than the one that survived, and they are listed here because a result selected out of 120 attempts needs its siblings on the record.

Component	Verdict	Evidence
9 EMA of session VWAP (1h trend filter)	Removed	Cut two thirds of trades; win rate rose without it, 35.62% vs 35.11%. Contributed nothing.
Breakout entries	Removed	Gross edge +0.049R, under half the fades. Trades against the only measured effect.
Order blocks	Removed	33.33% against a 33.33% benchmark, z −0.00, on 4,856 retracements. The coin flip to four decimals.

Fair value gaps	Kept	Cut retracements by 66% and the win rate rose 1.6 points. A location condition, not a depth one.
Move stop to breakeven at +10 pips	Removed	At 2:1 it must save more than two losers per winner killed. It does not.
GARCH / VaR / correlation regime gates	Removed	In-sample mean R rose from -0.049 to $+0.048$ across 24 configs; held-out -0.352 against -0.210 ungated. Fitted noise.
XAU/USD	Removed	1.62bp spread, $\sim 5\times$ the majors. $-0.058R$ despite $+0.090R$ gross.
News-time blackout	Removed	Measured spread at 08:30 and 14:00 ET is $0.8\text{--}1.0\times$ the daily median on the majors. The widening is sub-minute.

Two estimators that were checked against the truth and failed

Before the ask side was downloaded, the spread had to be estimated. Both standard approaches were implemented and both proved wrong at this frequency. **Corwin–Schultz** (2012), which infers spread from high–low ranges, called the 17:00 rollover — the widest hour of the day at $6\text{--}9\times$ — the *narrowest*; at five minutes the range is dominated by price movement, not bid–ask bounce. **Inverse tick activity** got the rollover right but priced news windows cheap, because ticks and spread spike together there. Both remain in the codebase with the numbers that condemn them.

6 Stability

A mean is only as good as its path. The quarterly record is the closest thing to a persistence test available inside the sample.

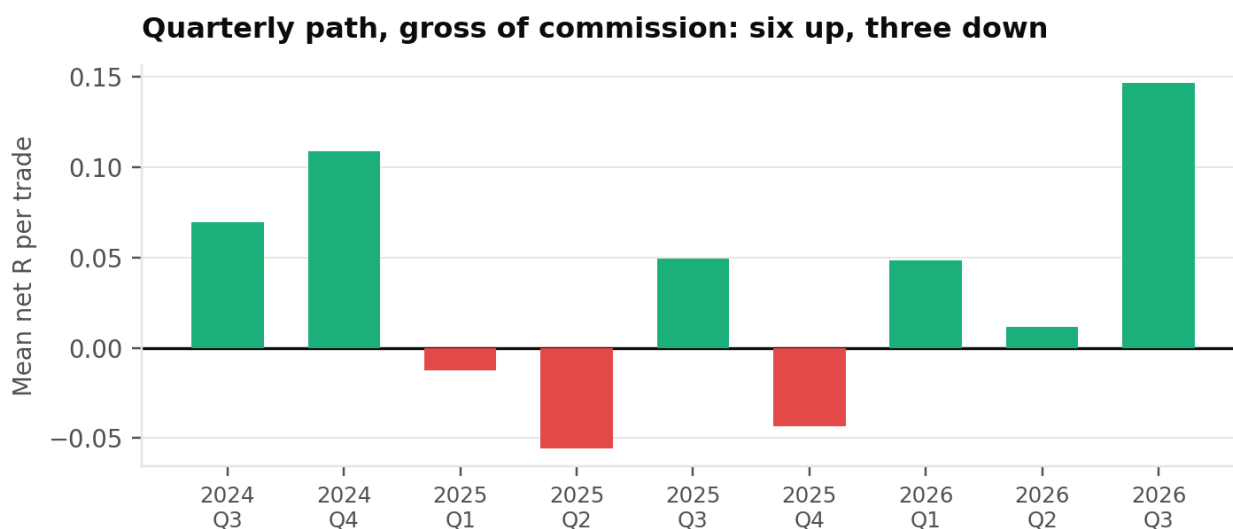


Figure 3. Mean net R per trade by quarter, net of the measured spread and gross of commission. Six quarters positive, three negative, worst quarter $-0.053R$. Earlier versions of this strategy had their entire mean sitting in two quarters out of eight; this one does not.

The honest caveat on all of it

This configuration was selected out of roughly 120 on *this* sample, and $t = +1.25$ is exactly the magnitude that selection alone can manufacture. One further finding makes the point concrete: longs outperformed shorts across the whole period by a wide and tempting margin — and splitting 95 instrument-months by each instrument's own return showed the advantage **reverses sign in down months**, with a correlation of $+0.45$ to the contemporaneous move and an intercept at zero return of $+0.011$. The long skew was the 2024–26 bull market,

not skill. Both sides were kept.

Whatever survives that kind of attack inside a sample still has to survive a period it has never seen.

7 Verdict

Where this lands

The effect is real. $z = +3.85$ against a correctly specified no-information benchmark, on 3,119 trades, stable across three instruments and six of nine quarters. Fading a failed extreme carries information in these markets.

The strategy is not tradeable. $+0.032R$ gross of commission at $t = +1.25$ is not significant, and $-0.109R$ at any realistic retail cost is unambiguous. Slippage would make it worse.

The constraint is frequency, not the rule. Every improvement so far has come from removing components, and the rule is now close to minimal. What remains is a small effect traded far too often to survive friction.

The next step is pre-registered. The frozen rule will be run once on September 2021–2023 — a window containing the 2022 Nasdaq bear market and gold's drawdown, which the rule has never seen. It passes only if the benchmark gap survives with the same sign. That test is written down before the data is fetched, and it will be reported whichever way it lands.

Method notes

Data. Dukascopy 5-minute bars, bid and ask, 22 Sep 2024 – 22 Sep 2026. Bars with zero ticks dropped; 4% single-bar jumps dropped. 30-minute and 1-hour frames are exact aggregations of the 5-minute bars, never resampled downward.

Exits. Triple-barrier. Where a bar touches both the stop and the target, the outcome is recorded as a *stop* — the pessimistic reading, since bar data cannot say which came first.

Overlap. Entries opening while a prior trade is live are dropped rather than stacked; counting each at one unit of risk would silently leverage the book.

Look-ahead. Every higher-timeframe value is taken from the last bar that had *closed*. Fair value gaps are indexed by their third bar and order blocks by the break that identifies them, never by the candle they formed on. Nine tests perturb future bars and assert that no earlier decision moves. The full suite is 241 tests.

Reproducibility. Code and results at github.com/mannankhanduja-max/FX-Quantitative-Risk-Engine.

Backtest research only. This document is not investment advice, and describes a strategy that lost money after realistic costs in every configuration tested.